II. Give very short answer to the following questions.

21. What is Inferential Statistics?
22. Define sample space.
23. What do you understand by random sample?
24. What are the different types of errors in hypotheses testing problem?
25. Prescribe the rejection rules for testing H_0: $\mu = \mu_0$ against all possible alternative hypotheses.
26. List the possible alternative hypotheses and the corresponding rejection rules followed in testing equality of two population means.
27. Specify the alternative hypotheses and the rejection rules prescribed for testing equality of two population proportions.
28. Define parameter.
29. What is statistic?
30. Define standard error of a statistic.
31. What do you understand by sampling distribution of a statistic?
32. What is null hypothesis?
33. Distinguish between null and alternative hypotheses.
34. For testing a null hypothesis against a two-sided alternative hypothesis, if $|z_0| > z_{\alpha/2}$ is found to hold, what will be your decision?
35. In a recent survey, conducted among 2000 randomly selected graduates in a district, 367 of them are IAS aspirants. Calculate the proportion of IAS aspirants among the graduates in the district.
36. Which type of test should be used for testing $H_0 : \mu = 27$ against $H_1 : \mu > 27$?
37. Formulate the null and alternative hypotheses for testing whether the average time required to XI standard students to complete a Chemistry laboratory exercise is less than 30 minutes.
38. If μ denotes the population mean, then find the critical value to be used for testing H_0: $\mu = 100$ against $H_1 : \mu < 100$ based on 250 observations at 5% level of significance.
39. The mean height of students studying X Standard in a school is 150 cms and the average height of 15 randomly selected students is 155 cms. Identify the parameter, statistic and their values from these information.
40. Calculate standard error of the sample mean, when sample mean is 100, sample size is 64 and population standard deviation is 24.
41. Find standard error of the sample proportion $p = 0.45$ when the population proportion is 0.5 and the sample size is 100.

III. Give short answer to the following questions.

42. Describe Decision Table.
43. What are type I and type II errors?
44. What do you mean by level of significance?
45. Explain one-tailed and two-tailed tests.
46. Explain critical value.

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